

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2016–17

Zoology**Paper No. ZOB-502 : Functional Anatomy of Chordates**

Time: 3 hours

Full Marks: 70

Note: Write your Roll No. at the top immediately on the receipt of this question paper. The figures in the right-hand margin indicate marks. Answer six questions including Question 1 which is compulsory.

Question 1. Answer the following parts:

- (a) State whether the following statements are True or False. Justify your answers in 1 or 2 sentences each: [4 × 2 = 8]

- (i) Tree like arborescent organs act as accessory respiratory organ in *Anabas*.
- (ii) Ceruminous glands are modified Sudoriferous glands.
- (iii) In Agnathans gill system also function as a food-collecting system.
- (iv) Dermatocranium forms the jaws in gnathostomes.

- (b) Define the following: [3 × 1 = 3]

- (i) Amphicoelous vertebra
- (ii) Mullerian duct
- (iii) Parabronchi.

- (c) Match the following: [6 × $\frac{1}{2}$ = 3]

Column–A		Column–B	
(i)	Meckel's cartilage	(1)	Solenodont condition
(ii)	Contour feather	(2)	Poison gland
(iii)	Ruminants	(3)	Ganoid scales
(iv)	Parotid gland	(4)	Pyloric caeca
(v)	Primitive bony fish	(5)	Quills
(vi)	Ray finned fish	(6)	Lower jaw
		(7)	Plumule
		(8)	Sebaceous gland

- (d) Differentiate between the following: [3 × 2 = 6]

- (i) Horns and Antlers
- (ii) Secodont condition and Lophodont condition
- (iii) Bipartite and Bicornuate uterus.

Question 2. With the help of neat labelled diagrams, explain different types of jaw suspensorium in vertebrates. [10]

Question 3. Give an illustrated account of gill system in shark. [10]

Question 4. What are aortic arches? Give an account of modifications in aortic arches from elasmobranchs to anurans with the help of neat labelled diagrams. [10]

Question 5. Explain the evolution of cerebellum in vertebrates. [10]

Question 6. Explain the following in brief: [2 × 5 = 10]

- (a) Dentition in mammals
- (b) Hepatic portal system.

Question 7. Give an account of different types of kidney tubules in vertebrates. [10]

Question 8. Write short notes on the following: [2 × 5 = 10]

- (a) Digital cornifications
- (b) Mesonephros kidney.

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